

SAACHI SWARUP BHOSALE

Computer Science Engineering Student

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SUMMARY

Final-year Computer Science Engineering student at BITS Pilani, Dubai (Minor in Data Science), with research and project experience spanning cybersecurity risk modeling, machine learning, and automation. Co-author of a conference paper on graph-based cyber defense prioritization currently under review for publication, and researching Meta Attack Language design for healthcare cyberinfrastructure. Proven ability to build end-to-end applications from ML pipelines to workflow automation tools and eager to contribute technical and analytical skills to a professional engineering team.

EXPERIENCE

Data Analyst Intern

Casamia

06/2025 – 08/2025

Dubai

On-site Internship

- Built and maintained a sister company website for casamia, including front-end development and content integration.
- Performed web scraping to collect and structure external data for business use.
- Automated business workflows using Microsoft Power Apps, Power Automate, and n8n, reducing manual data-entry effort.
- Developed a machine-learning-driven PowerPoint generator application using Flask, HTML, and SQL, enabling automatic creation of presentations from structured inputs.
- Implemented AI-powered data extraction to streamline information capture and internal reporting.

EDUCATION

Bachelor of Engineering in Computer Science (Minor in Data Science)

BITS Pilani, Dubai Campus

Current CGPA: 7.40

2023 – Present

Dubai

CBSE Class XII

GEMS Our Own English High School, Dubai

Score: 92.3%

2023

Dubai

CBSE Class X

GEMS Our Own English High School, Dubai

Score: 89%

2021

Dubai

RESEARCH & PUBLICATIONS

Improving Risk Analysis and Defense Prioritization in Cybersecurity Using Enhanced PageRank and Knapsack Optimization

Submitted for Publication

- Co-developed a modified PageRank algorithm combined with knapsack-based cost optimization to rank and select cybersecurity defense actions under budget constraints.
- Modeled real-world attack graphs and validated the approach through simulation, showing measurable gains in risk reduction over traditional prioritization methods.

Design of Meta Attack Language for Healthcare Cyberinfrastructure

Ongoing Research

- Researching a domain-specific Meta Attack Language (MAL) to formally model threats, vulnerabilities, and attack paths across healthcare IT and medical device infrastructure.
- Investigating how MAL-based threat modeling can support proactive risk assessment for critical healthcare systems.

TECHNICAL SKILLS

Languages & Databases

Java, Python, C, C++, JavaScript, HTML, CSS, SQL

AI & Automation

OpenAI API, Machine Learning, Power Automate, n8n, Power Apps, Git/GitHub, API & Security

Productivity Tools

MS Word, Excel, PowerPoint, SharePoint

Data & ML

Scikit-learn, Pandas, NumPy, Random Forest, K-Means, SVM

Cybersecurity

AES/E2E Encryption, Attack Graph Modelling, Threat Prioritisation, ClamAV

PROJECTS

Budget-Constrained Defense Prioritization Using PageRank-Based Attack Graph Analysis

- Developed a Python framework applying a modified PageRank algorithm and cost-aware optimization to prioritize cybersecurity defenses under limited budgets.
- Simulated attack graphs to demonstrate improved resource allocation and decision-making versus traditional approaches. Tools: Python, NetworkX, cybersecurity risk modeling.

Cryptographic Access-Control Framework for Secure File Operations

- Designed a framework combining cryptographic techniques with access-control policies to secure file read/write/share operations.
- Implemented encryption and permission-based validation to protect sensitive data against unauthorized access.

F1 Race Outcome Predictions

- Built an ML pipeline using multi-season Formula 1 data for grid and race outcome prediction, covering preprocessing, feature engineering, and hyperparameter tuning.
- Evaluated models using MAE, RMSE, and F1-score, delivering insights on key performance factors.

Hybrid Cluster-Routing Classifier for Academic Stress Prediction,

- Designed a two-stage mixture-of-experts pipeline: K-Means segments 1,100 student records into behavioural profiles; a dedicated specialist classifier per cluster delivers the final stress prediction.
- Achieved 0.8729 macro-F1, outperforming Decision Tree, k-NN, Naive Bayes, and SVM baselines by up to 2.7 percentage points.
- Engineered 5 composite psychometric features from 20 raw survey scales; validated rigorously using nested cross-validation, McNemar's test, and bootstrap confidence intervals.
- Deployed as a Streamlit web application, enabling universities to flag at-risk students in real time and trigger targeted counselling or workload intervention.

Smart Ride Demand Prediction Using Machine Learning and Transfer Learning

- Building a demand-forecasting model for ride-hailing services, applying transfer learning to improve prediction accuracy across regions with limited data.

PPT Generator Web App

Created a Flask-based tool that automatically generates PowerPoint presentations from user inputs, streamlining content-creation workflows.

Airline Management System

Implemented a console-based Java system using OOP principles for flight scheduling, bookings, and ticket management with modular, scalable architecture.

ADDITIONAL INFORMATION

Soft Skills: Effective communication, problem solving, time management, adaptability and continuous learning
Languages: English, Hindi, Marathi